Economic Forecasting
Advice on Empirical Project

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Below, we list some general advice and tips on empirical projects.

1. **Scientific writing**: the structure and good scientific writing in the project is important.

2. **Methods**: a typical empirical project can be characterized as a "horse race" between several methods. Make sure to explain the underlying methodology in a self-contained manner. Additional attention should be given to methodologies that are out of the course curriculum (e.g. machine learning or new variants of curriculum methods). When introducing methods, it is beneficial if students' own insights are also discussed.

3. **Initial analysis of data**: e.g. what are the properties of the data (stationarity, seasonality, etc) and how does this influence the forecasting task?

4. **Forecasting setup**: make sure to setup the forecasting exercise in a "correct" way, where "correct" may depend on the forecasting application in question. For example, in-sample and out-of-sample periods are explicitly mentioned, and make sure that forecasts don't use forward-looking information when estimating the models.

5. **Discussion of choices**: you need to make many choices in the project, e.g. on list of methods, start and end of data set, in- and out-of sample split, rolling vs expanding window estimation, forecast evaluation method. You need to include discussions about reasons for these choices, and in some cases also brief discussions about the excluded options.

6. **Evaluation of forecasts**: in most cases, you should go beyond just reporting a number (e.g mean squared error) on forecasts. In particular, you should (if possible) assess statistical significance of your results (e.g. through a Diebold-Mariano test, a Clark-West test, encompassing tests, Model Confidence Set, etc). Including other methods for evaluating forecasts would be beneficial (e.g. economic measures of performance).

7. **Empirical section**: Since this is a course in forecasting, the empirical analysis section on the forecasting accuracy of the methods should account for the bigger part of the final project text.

8. **Literature review**: which articles/books in the literature are related to the research question and methodology of the empirical project? In most cases, reference list includes 10+ articles.

9. **Appropriate level**: this is a 5000 level course and hence projects typically go beyond what you have learned in the bachelor curriculum. Your supervisor can provide feedback in adjusting the level correctly. This can be done in various ways, e.g. including at least one of: (i) new methods, (ii) larger/interesting dataset, (iii) deeper empirical analysis and very careful execution of methods.

Disclaimer: Although this list covers many issues relevant to empirical projects, it should not be read as a formal assessment tool.